ADDITIONAL RESULTS ON OTHER REACTION EQUILIBRIA

Note on Updates

Future updates or revisions to this document will be made available at: https://github.com/daferro/Notebooks-Chemical_Equilibrium

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1. What is this document about?

This document summarizes representative results obtained with ***Notebook 2*** when applied to gas-phase reactions beyond the $\mathbf{N_2O_4 \rightleftharpoons 2NO_2}$ case study. The purpose is twofold: (i) to illustrate the generality of the notebook workflow and (ii) to provide reference outputs that can be used for comparison and teaching. In particular, we analyze the following equilibrium:

- $N_2 + 3\ H_2 \rightleftharpoons 2\ NH_3$
- $O_2 + 2\ H_2 \rightleftharpoons 2\ H_2O$

2. The $N_2 + 3H_2 \rightleftharpoons 2NH_3$ equilibrium

When running ***Notebook 2***, this reaction is defined by the following input string:

- `N2 + 3 H2 -> 2 NH3`

The remaining input parameters required by the notebook are listed in Table 1.

Table 1. Input data for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction. $\Delta_f H^\circ$ is reported in kJ·mol⁻¹, and S°_m and $C^\circ_{p,m}$ in J·mol⁻¹·K⁻¹. These three values correspond to 298 K.[1]

	$\Delta_f H^\circ$	S°_m	$C^\circ_{p,m}$	PubChem CID	Molecular charge	Number of unpaired electrons	Symmetry number
N_2	0.00	191.61	29.125	947	0	0	2
H_2	0.00	130.684	28.824	783	0	0	2
NH_3	-46.11	192.45	35.06	222	0	0	3

For initial amounts equal to the stoichiometric coefficients of the reactants (*i.e.* 1 mol of N_2 and 3 of H_2, and 0 mol of NH_3), the variation of $G(\xi)$ and of $A(\xi)$ with the extent of reaction, ξ, is shown in Figs. 1 and 2, respectively.

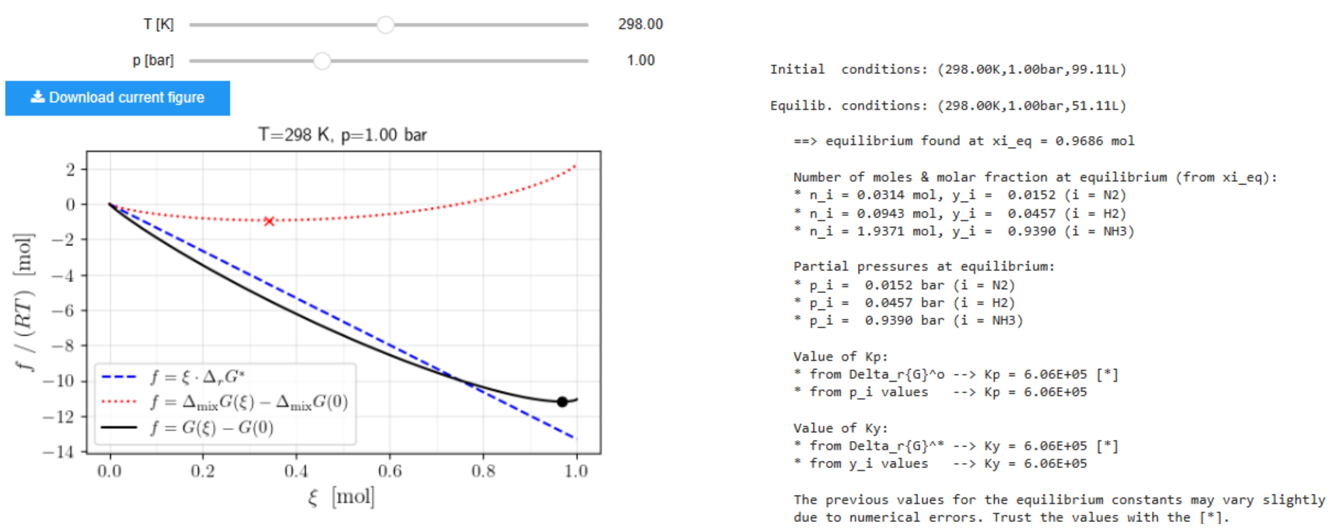

Figure 1. Screenshots showing equilibrium results at constant p and T for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction. Moving the sliders updates the plot interactively.

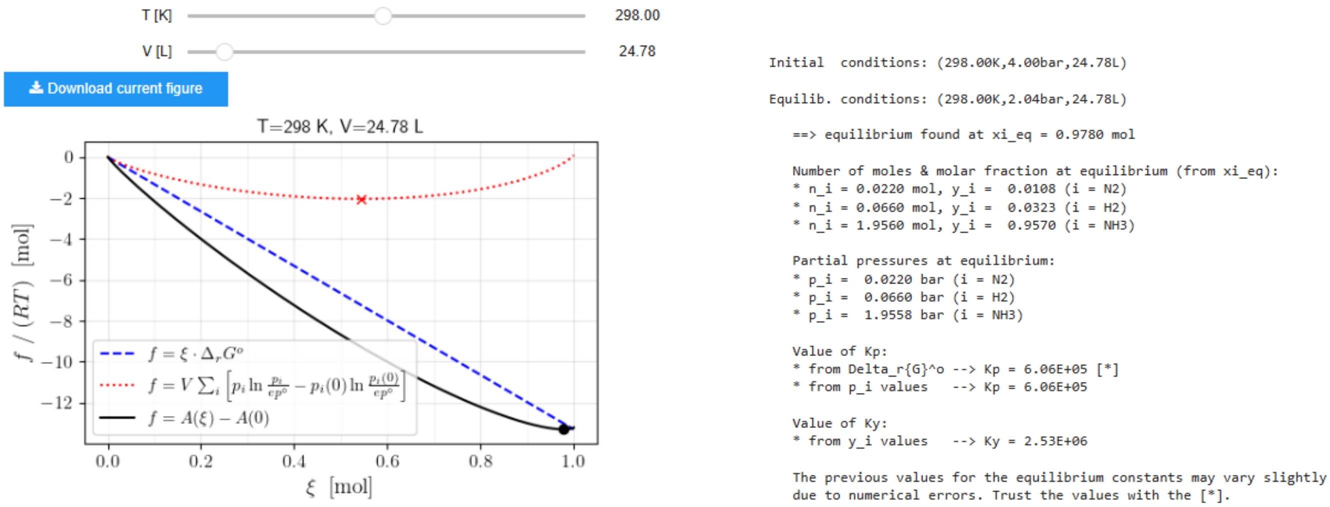

Figure 2. Screenshots showing equilibrium results at constant V and T for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction. Moving the sliders updates the plot interactively.

In the *Statistical-mechanical* section of ***Notebook 2***, the user provides the **PubChem CID** values in order to retrieve initial geometries for the reactants and products. The CIDs used in this example are listed in Table 1. An illustrative output is shown in Fig. 3.

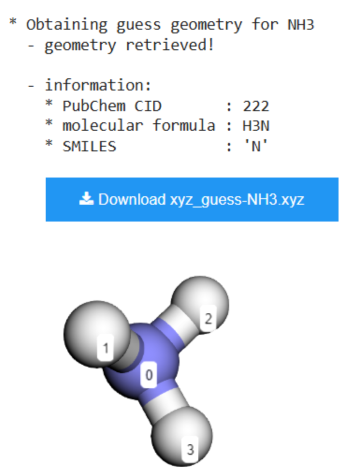

Figure 3. Screenshot showing the retrieved initial structure for NH_3. The view is interactive, allowing the molecule to be inspected from different angles.

For this system, we recommend using the **PBE0** functional and the **6-31G*** basis set. This combination was selected based on benchmarking against experimental data.[1, 2] The vibrational frequencies associated with the optimized geometries are listed in Table 2.

Table 2. Comparison of experimental[a] and theoretical vibrational frequencies (cm^{-1}) for H_2, N_2 and NH_3

	N_2	H_2	NH_3
Experimental[a]	2359	4401	950, 1627 (2), 3337, 3444 (2)
PBE0/6-31G*	2498	4451	1131, 1733 (2), 3486, 3627 (2)

[a]Experimental data taken from NIST Chemistry WebBook.[2]

After correcting the symmetry numbers according to Table 1, the thermodynamic quantities reported in Table 3 are obtained.

Table 3. Comparison of experimental[a] and theoretical values of $\Delta_r H°$, $\Delta_r S°$, $\Delta_r G°$ for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction at 298 K

	$\Delta_r H°$ ($kJ \cdot mol^{-1}$)	$\Delta_r S°$ ($J \cdot mol^{-1} \cdot K^{-1}$)	$\Delta_r G°$ ($kJ \cdot mol^{-1}$)
Experimental[a]	-92.22	-198.8	-32.99
PBE0/6-31G*	-90.99	-198.0	-32.00

[a]Experimental data taken from Atkins' *Physical Chemistry* book.[1]

In addition, the temperature dependence of $\Delta_r G°$ shown in Fig. 4 is obtained. Notice that it behaves according to the classical thermodynamics expression, given by Eq. (2) of the main manuscript, if $\Delta_r C_p°$ is set to $-5.71 \cdot R$.

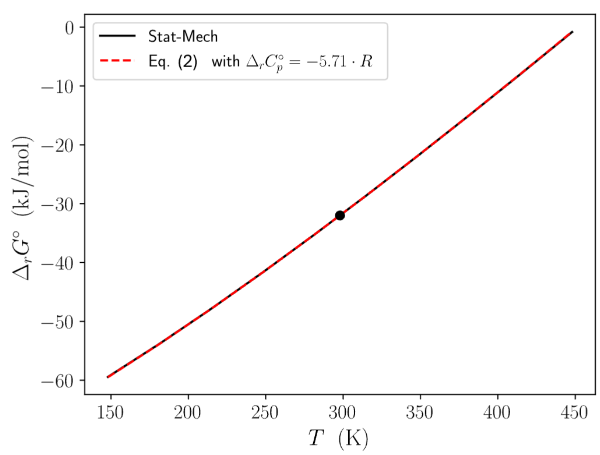

Figure 4. Standard Gibbs energy of reaction for $N_2 + 3H_2 \rightleftharpoons 2NH_3$ as a function of T. Statistical-mechanical results (**black** solid) are compared with the classical thermodynamics expression (**red** dashed) computed using PBE0/6-31G* values of $\Delta_r H°$ and $\Delta_r S°$ at T_{ref} =298 K. A fitted value $\Delta_r C_p° = -5.71 \cdot R$ is used.

3. The $O_2 + 2H_2 \rightleftharpoons 2H_2O$ equilibrium

When running ***Notebook 2***, this reaction is defined by the following input string:

- `O2 + 2 H2 -> 2 H2O`

The remaining input parameters required by the notebook are listed in Table 4.

Table 4. Input data for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction. $\Delta_f H^\circ$ is reported in kJ·mol^{-1}, and S°_m and $C^\circ_{p,\mathrm{m}}$ in J·mol^{-1}·K^{-1}. These three values correspond to 298 K.[1]

	$\Delta_f H^\circ$	S°_m	$C^\circ_{p,\mathrm{m}}$	PubChem CID	Molecular charge	Number of unpaired electrons	Symmetry number
O_2	0.00	205.138	29.355	977	0	2	2
H_2	0.00	130.684	28.824	783	0	0	2
H_2O	-241.82	188.83	33.58	962	0	0	2

For initial amounts equal to the stoichiometric coefficients of the reactants (*i.e.* 1 mol of O_2 and 2 of H_2, and 0 mol of H_2O), the variation of $G(\xi)$ and of $A(\xi)$ with the extent of reaction, ξ, is shown in Figs. 5 and 6, respectively.

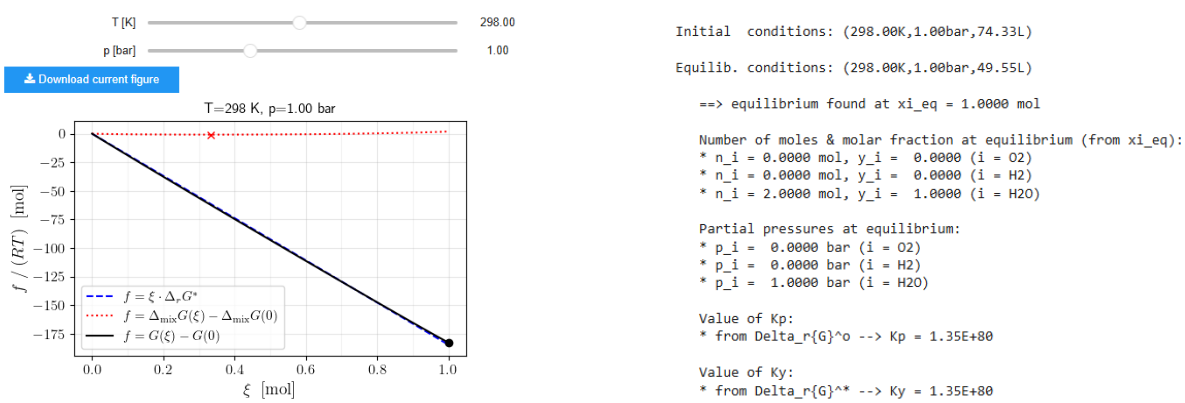

Figure 5. Screenshots showing equilibrium results at constant p and T for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction. Moving the sliders updates the plot interactively.

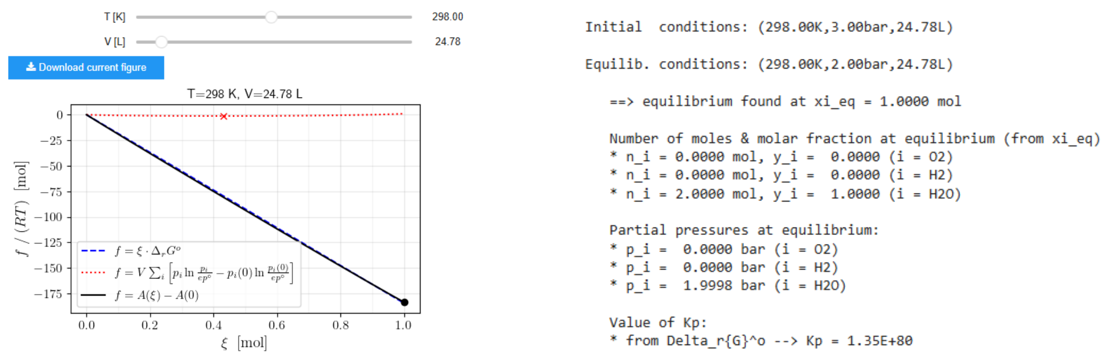

Figure 6. Screenshots showing equilibrium results at constant V and T for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction. Moving the sliders updates the plot interactively.

In the *Statistical-mechanical* section of ***Notebook 2***, the user provides the **PubChem CID** values in order to retrieve initial geometries for the reactants and products. The CIDs used in this example are listed in Table 4. An illustrative output is shown in Fig. 7.

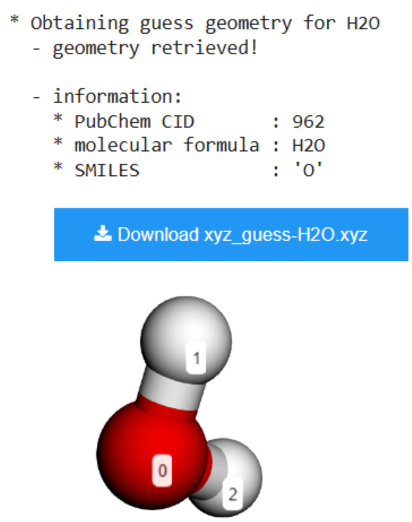

Figure 7. Screenshot showing the retrieved initial structure for H_2O. The view is interactive, allowing the molecule to be inspected from different angles.

For this system, we recommend using the **M06-2X** functional and the **aug-cc-pVDZ** basis set. This combination was selected based on benchmarking against experimental data.[1, 2] The vibrational frequencies associated with the optimized geometries are listed in Table 5. Note that electronic-structure calculations for O_2 must be performed with the **correct open-shell configuration** (two unpaired electrons).

Table 5. Comparison of experimental[a] and theoretical vibrational frequencies (cm^{-1}) for H_2, O_2 and H_2O

	O_2	H_2	H_2O
Experimental[a]	1580	4401	1595, 3657, 3756
M06-2X/aug-cc-pVDZ	1771	4477	1614, 3861, 3971

[a]Data taken from NIST Chemistry WebBook.[2]

After correcting the symmetry numbers according to Table 4, the thermodynamic quantities reported in Table 6 are obtained.

Table 6. Comparison of experimental[a] and theoretical values of $\Delta_r H°$, $\Delta_r S°$, $\Delta_r G°$ for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction at 298 K

	$\Delta_r H°$ (kJ·mol^{-1})	$\Delta_r S°$ (J·mol^{-1}·K^{-1})	$\Delta_r G°$ (kJ·mol^{-1})
Experimental[a]	-483.6	-88.64	-457.2
M06-2X/aug-cc-pVDZ	-477.7	-88.58	-451.3

[a]Data taken from Atkins' *Physical Chemistry* book.[1]

In addition, the temperature dependence of $\Delta_r G°$ shown in Fig. 8 is obtained. Notice that it behaves according to the classical thermodynamics expression, given by Eq. (2) of the main manuscript, if $\Delta_r C_p°$ is set to -2.47·R.

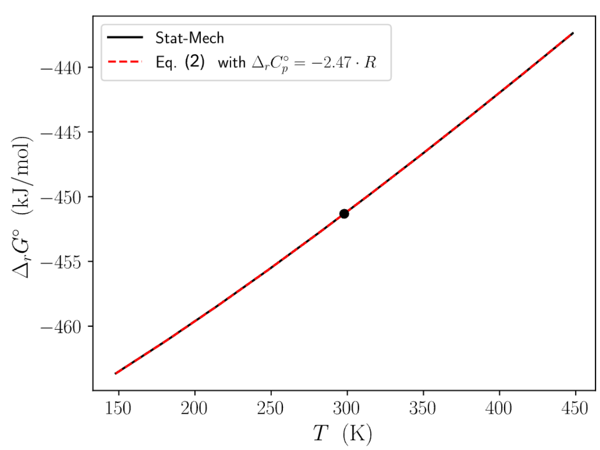

Figure 8. Standard Gibbs energy of reaction for $O_2 + 2H_2 \rightleftharpoons 2H_2O$ as a function of T. Statistical-mechanical results (**black** solid) are compared with the classical thermodynamics expression (**red** dashed) computed using M06-2X/aug-cc-pVDZ values of $\Delta_r H°$ and $\Delta_r S°$ at T_{ref} =298 K. A fitted value $\Delta_r C_p° = -2.47 \cdot R$ is used.

4. References

(1) Atkins, P. W.; De Paula, J.; Keeler, J. *Atkins' Physical Chemistry*, 11th ed.; Oxford University Press, London, 2018.

(2) Linstrom, P. J.; Mallard, W. G. The NIST Chemistry WebBook: A Chemical Data Resource on the Internet. *J. Chem. & Eng. Data* **2001**, *46* (5), 1059-1063. DOI: 10.1021/je000236i.